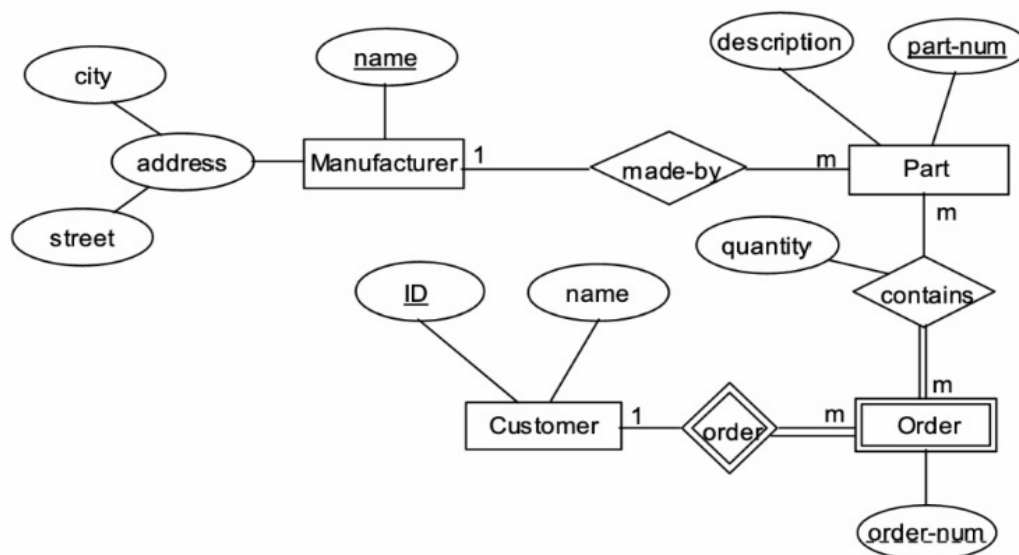


1. What is the purpose of data base system?
2. Describe the three levels of data abstraction?
3. Describe schema and instance in DBMS.
4. What are the different types of schema?
5. Explain the database design process in detail.
6. Explain the difference between meta data and data dictionary with example.
7. What are the different data models?
8. Define Entity, Attributes, Entity set, relationship with appropriate notations in ERD?
9. Draw the notation for multivalued attributes? Give one example?
10. Discuss about Database users and Administrators?
11. Draw ER diagram for Ternary Relationship set with suitable example? What does the cardinality ratio specify?
12. List any eight applications of DBMS.
13. Explain about Database languages with examples?
13. Define Relational Database. Explain data representation in RDBMS.
14. Convert logical design (as in given figure) to physical database design.



15. Draw the Architecture of Database?
16. What are the two types of participation constraint?
17. What are different access types in DML?
18. Differentiate File Processing System with Database Management System.
19. Explain about integrity constraints over relations?
20. Draw the ER diagram for a company needs to store information about employees(Identified by ssn, with salary and phone as attributes), departments(Identified by

eno, with ename and budget as attributes), and children of employees (With name and age as attributes). Employees work in departments, each department is managed by an employee, a child must be identified uniquely by name when the parent (Who is an employee; assume that only one parent works for the company) is known. We are not interested in information about a child once the parent leaves the company.

21. Explain the terms with example: Weak Entity set, candidate key, primary key, foreign key

22. List and Explain set operators of relational algebra.

23. State the difference between cartesian product and natural join in context of relational algebra.

24. Give an example of how to use the rename operation in relational algebra.

25. Explain the following

- i. Data Definition Language.
- ii. Data Manipulation Language.
- iii. Data control Language.
- iv. Views

26. What are aggregate functions? Explain five built-in aggregate functions.

27. What is a view? How can it be created? Explain with an example.

28. What is a join operator? Explain about conditional join and natural join with syntax and example.

29. How can we compare using null values? Explain about logical connectives with examples.

30. Write about triggers in databases with example.

31. Explain the structure of basic form of an SQL query.

32. What is cursor in database? How can you create an user defined cursor to store all manager details in SQL?

33. Consider the given relations:

STUDENT(SID, Name, Dept, Age)

COURSE(CID, CName, Dept)

ENROLL(SID, CID, Marks)

Write Relational Algebra expression for the following queries:

Find names of all students.

Find names of students whose age is greater than 20.

Find students enrolled in DBMS.

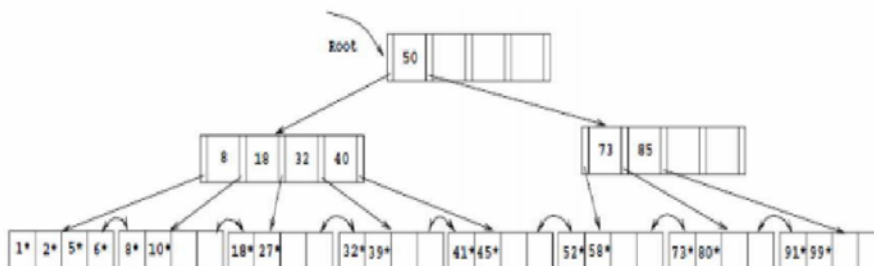
Find students who are not enrolled in any course.

Find students who are enrolled in all courses.

34. What is redundancy? Explain with suitable example.

35. What are the anomalies created by data redundancy? Explain.

36. Explain the classification of functional dependency.
37. Consider the relation schema $R(A,B,C,D,E,F)$ and the functional dependencies $A \rightarrow B, C \rightarrow DF, AC \rightarrow E, D \rightarrow F$. Find the candidate keys of this relation R ?
38. Define and explain 4NF with suitable example.
39. Explain join dependencies with fifth normal form with an example.
40. What is meant by multivalued dependency? Explain with example.
41. Consider the relation $R(v,w,x,y,z)$ with functional dependencies $\{z \rightarrow y, y \rightarrow z, x \rightarrow y, x \rightarrow v, vw \rightarrow x\}$.
42. Find the closure for $\{xy\}$ given functional dependencies.
43. Define the terms in the context of file and storage management. Seek time, access time, rotational latency.
44. Compare Primary indexing and clustering indexing in file management techniques.
45. Disadvantages of B TREE over B+ TREE
46. Consider the following relation: Shipping (Shipname, Shiptype, Voyageid, Cargo, Port, Date) (note: Date the date the ship arrives in the given port) with functional dependencies- $\{ShipName \rightarrow Shiptype, Voyageid \rightarrow Shipname, Cargo, Shipname, Date \rightarrow Voyageid, Port\}$
- a) Identify the candidate keys.
- b) Normalize the relation until it is completely normalized.
47. Define BCNF. How does it differ from 3 NF ? Why is it considered as stronger than 3 NF ?
48. Discuss Armstrong's Axioms.
49. Explain, how secondary indexing can help in join operation.
50. Given below are two sets of FD's for a relation $R(A,B,C,D,E)$. Are they equivalent?
 $F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$ and $G = \{A \rightarrow CD, E \rightarrow AH\}$
51. "All BCNF relations are in 3 NF but not vice versa." Prove the statement with example.



52. Consider insertion sequence- 2,3,5,11,17,19,23,29,31. Construct B+ Tree for the case where the number of pointers that will fit in one node is 4.

53. Draw the precedence graph for the schedule w1 (A) w2 (A) r3 (B) w3 (B) r1 (B):
S:{r1(x);r2(y);r3(y);w2(y);w1(x);w3(x);r2(x);w2(x)}

Test whether the schedule is conflict serializable or not? If Yes, then what is the correct order of serialization.

54. Define ACID properties of a transaction in DBMS.

55. Discuss different states of transaction with diagram.

56. Explain serial schedule with example.

57. When a schedule is called non-recoverable schedule? Explain.

58. Define Lock. Define two-phase locking.

59. Differentiate between Serial Schedule and serializable schedule. Explain different types of locks.

60. Explain the concept of Hash File Organization with suitable diagram.

61. Compare ordered indexing and hashing. How does multilevel indexing improve the efficiency of searching an index file?

62. Given a block can hold either 3 records or 10 key pointers. A database contains n records, then how many blocks do we need to hold the data file and the dense index.

63. The order of a leaf node in a B+ tree is the maximum number of (value, data record pointer) pairs it can hold. Given that the block size is 1K bytes, data record pointer is 7 bytes long, the value field is 9 bytes long and a block pointer is 6 bytes long, what is the order of the leaf node?

64. What is referential integrity constraints? How it is related to foreign key. Explain with example.

65. Discuss Three Tier Database Architecture with diagram.

66. Create ER diagram for each of the followings

1. Each company operates four departments and each department belongs to one company.
2. Each department in part (a) employs one or more employees and each employee works for one department.
3. Each of the employees in part(b) may or may not have one or more dependents.

67. What are the differences between natural join and cross product of two relations? Explain with example.

68. Discuss each of followings- (a)Natural Join b) equi-join c) self join d) inner-join e)outer join

69. Consider the following relation schema

Works(Pname,Cname,salary)

Lives(Pname,Street,City)

Located_in (Cname, city)

Manager(Pname,Mgrname)

Write the SQL queries for the following.

- i) Find the names of all persons who live in the city Bangalore.
- ii) Retrieve the names of all person of "Infosys" whose salary is between Rs .50000
- iii) Find the names of all persons who lives and work in the same city
- iv) List the names of the people who work for "Tech M" along with the cities they live in.
- v) Find the average salary of "Infosys" persons